TerraFlow BioAI

FINAL MAJOR PROJECT REPORT

A Zone-Aware Embedded Agro-Environmental Monitoring and Advisory System for Precision Irrigation and Crop Management

Final project identity

One physical node (TF01) manages one defined management zone (Z01). Irrigation is the only physically automated agricultural function. Crop suitability and fertilizer functions are advisory. STM32 remains the local safety and actuation authority.

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Project type: 4-credit academic major project; research publication is a secondary goal

Report status: Final consistency version based on the locked architecture and roadmap

Academic year: _____

Core story: Sense -> Understand -> Advise -> Automatically Irrigate -> Verify

Document. Control and Final Consistency Lock

This report consolidates the supplied Phase 1-4 roadmap, detailed project report, complete project guide, the final architecture image, and the latest decisions agreed for the project. Older draft details are not allowed to override the final locked scope.

Item	Final decision
Node / zone	One physical node TF01 and one experimental management zone Z01
Local controller	STM32F103C8T6
Connectivity	ESP32 over UART + Wi-Fi
Data / backend	Node.js / Express + Supabase + React dashboard
Preprocessing	1-D Kalman filter for soil-moisture signal as a distinct preprocessing block
Core sensors	Capacitive moisture, DS18B20, TDS, BME280, BH1750
Location	NEO-6M GPS
Final measurement / safety inputs	Flow sensor and float switch are mandatory final hardware additions
Agronomic knowledge base	Stored inside Supabase as structured reference/configuration tables
ML	Separate ML support layer; Random Forest retained as decision support; hybrid use is allowed, model selection remains evidence-driven
Irrigation control	Event/need-based target-volume control; no fixed 5-minute irrigation cycle
Water measurement	Flow sensor provides measured rate / cumulative volume for runtime and delivery verification
Crop advisory	Top 5 suitability recommendations with score and reasons; not a direct profitability prediction
Fertilizer advisory	Soil-test/reference N/P/K pathway; rule-based core with ML support to be analyzed and validated
Safety rule	Cloud/ML may recommend -> STM32 checks local conditions -> pump actuates -> feedback validates

Important status note

Phase 1 and Phase 2 are treated as completed foundations. Phase 3 and Phase 4 are the current/final implementation and validation stages. This report does not invent results that have not yet been measured.

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> **How to use this report**
>
> Read Chapters 1-3 first to understand the whole system. Chapters 4-13 explain how the system actually works. Chapters 14-17 explain what changed phase by phase. Chapters 18-22 show how to test, present and defend the work. The appendices are for implementation and viva revision.

1. Executive Summary and Mental Model

The project in one story

TerraFlow BioAI is a practical 4-credit embedded agricultural system. It combines a real sensor-and-pump node with a connected software layer that adds agricultural context. The project is deliberately

smaller than a commercial farm platform: one node, one zone, a limited agronomic dataset, and three clearly separated agricultural functions.

The three functions are simple to remember. Irrigation is the physical function: the system observes moisture, uses crop/soil/weather context, decides whether irrigation is needed, and automatically operates the pump under local STM32 safety rules. Crop suitability is a pre-planting or seasonal advisory: the software compares zone conditions with the crop library and returns the top five recommendations with reasons. Fertilizer is a separate advisory pathway: it uses entered or reference soil-test values and crop-stage information to produce a nutrient recommendation without pretending that the TDS sensor measures NPK.

The final system in one picture

```
FIELD SENSE -> STM32 -> KALMAN PREPROCESSING -> ESP32 -> BACKEND / SUPABASE
                                  |
                        WEATHER + AGRONOMIC CONTEXT
                                  |
                    ML SUPPORT + DECISION ENGINES
                      /           |           \
                IRRIGATION     CROP TOP 5     FERTILIZER
                    |
              STM32 SAFETY CHECKS
                    |
             PUMP + FLOW SENSOR
                    |
              ZONE Z01 FIELD
                    |
             MOISTURE FEEDBACK
```

> **The one sentence to memorize**
>
> TerraFlow senses a management zone, interprets the condition using crop, stage, soil and weather context, provides agricultural advice, lets the STM32 enforce local safety and automatic irrigation, and verifies the physical response.

1.1 What TerraFlow is not

- It is not a whole-farm sensing system. One node represents the defined zone Z01, not the entire farm automatically.
- It is not a direct NPK sensing system. Fertilizer uses soil-test/reference inputs.
- It is not an AI-only controller. The STM32 remains the physical safety and actuation authority.
- It is not a universal fertilizer or crop calculator. Advice is bounded by the configured knowledge base and chosen methods.
- It is not a fixed-timer irrigation system. The final irrigation event is driven by need and target water volume.

2. Problem, Motivation, Objectives and Scope

2.1 The practical problem

Fixed-schedule irrigation can water plants when water is not needed and can fail to react to changing soil conditions. A single moisture threshold also ignores the fact that different crops, soils, growth stages and weather situations behave differently. Crop choice and fertilizer decisions are separate agricultural questions that cannot be solved from the moisture sensor alone.

2.2 Project objective

To design and demonstrate a low-cost embedded agricultural system that continuously monitors soil and environmental conditions, automatically controls irrigation for a defined management zone, and provides crop-suitability and soil-test-based fertilizer advisories through a connected software platform.

2.3 Secondary research objective

Generate credible data and validation evidence that can later support one or two research papers. The publication goal remains secondary to completing the major project.

2.4 Locked scope

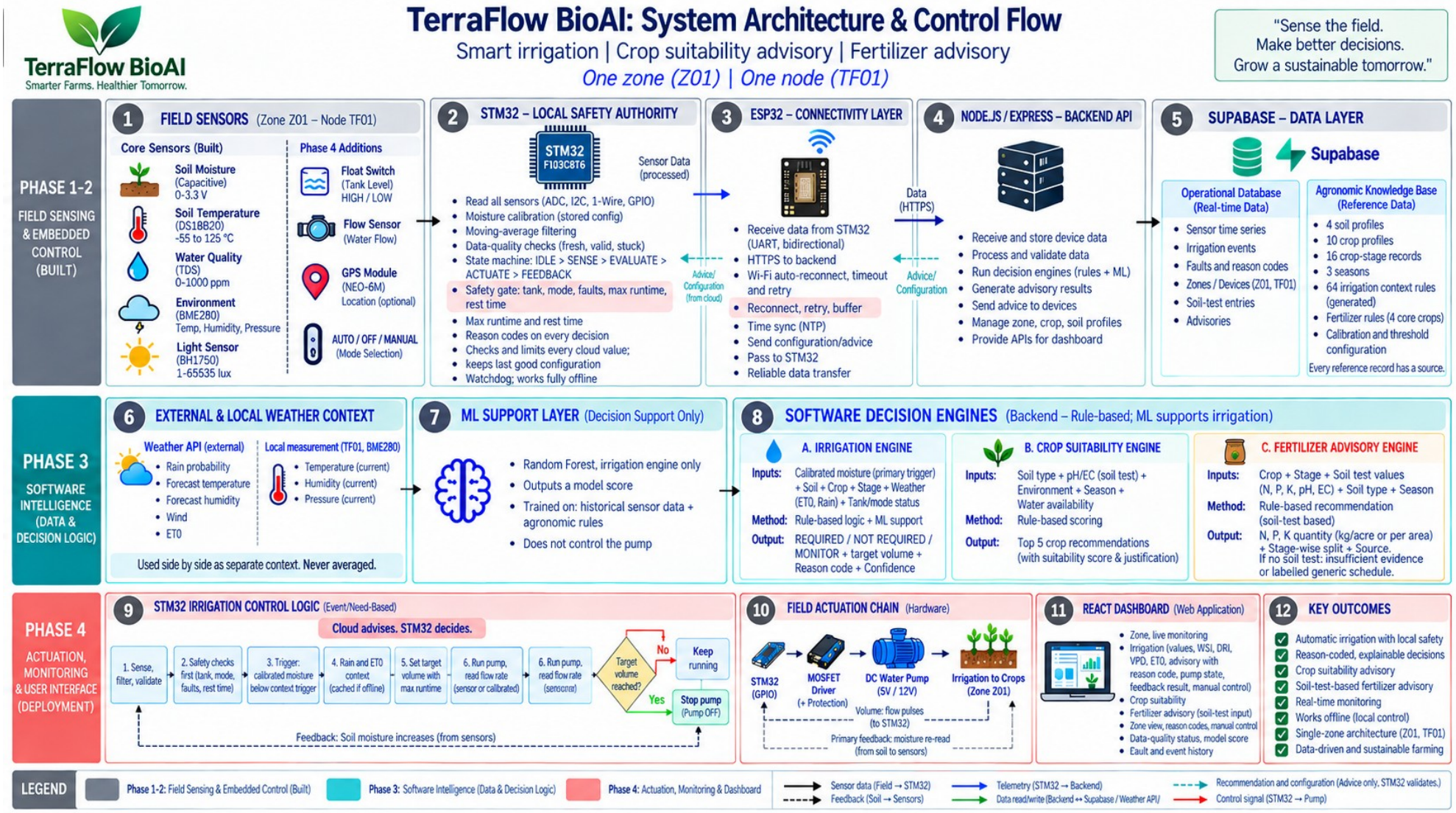
Dimension	Final scope
Physical deployment	1 node TF01 + 1 management zone Z01
Soil profiles	4 broad software/reference profiles: Black, Red, Alluvial/Muddy, Sandy/White
Crop profiles	10 crops: Rice, Wheat, Maize, Cotton, Soybean, Groundnut, Chickpea, Tomato, Onion, Potato
Detailed crop logic	Rice, Wheat, Maize, Tomato
Growth stages	Initial, Development, Mid-season, Late-season
Seasons	Kharif, Rabi, Zaid
ML	Random Forest retained as decision support; hybrid support layer for decision engines; exact fertilizer model remains an evaluation item
Agricultural functions	Automatic irrigation; crop suitability advisory; fertilizer advisory

2.5 Zone meaning

A management zone is the area intentionally managed together by one irrigation setup and represented by the TerraFlow node for this prototype. The software tags records with farm_id, zone_id and device_id. The zone abstraction can scale later, but multi-node deployment is not part of the current physical build.

3. Final System Architecture - Updated Diagram

Use this page as the master visual reference



3. Final System Architecture - Layer by Layer

3.1 Layer 1 - Field sensing in Z01

The field layer contains the five core environmental/water-quality sensors plus GPS and final water/safety instrumentation. The capacitive moisture sensor is the primary irrigation observation. DS18B20 provides soil/root-zone temperature. TDS provides dissolved-solids and irrigation-water quality context. BME280 provides local air temperature, humidity and pressure. BH1750 provides illumination. NEO-6M provides location metadata and supports location-based weather retrieval. The flow sensor measures water delivery; the float switch protects the pump from a low tank condition.

Input	Final role	Important limitation
Capacitive moisture	Primary irrigation variable	Must be calibrated for the demonstration soil; raw percentage is not universal VWC
DS18B20	Soil/root-zone temperature and context	Not a standalone pump trigger
TDS	Irrigation-water dissolved-solids / salinity-related context	Not NPK; does not produce fertilizer quantities
BME280	Local temperature, RH, pressure; VPD/WSI/DRI context	Current local state, not a forecast
BH1750	Illuminance / light context	Lux is not direct PAR
NEO-6M GPS	Zone location and weather lookup	Does not prove whole-farm coverage
Flow sensor	Measured flow rate and cumulative water volume	Requires sensor-specific calibration
Float switch	Tank-level safety input	Used to block pump when tank is low

3.2 Layer 2 - STM32 local control

The STM32F103C8T6 is the embedded authority. It reads sensors, performs validation and local safety logic, communicates over UART, controls the pump driver, monitors fault conditions, and remains capable of basic local irrigation even if the network is unavailable. The most important architectural rule is that the cloud does not own the pump.

3.3 Layer 3 - Kalman filter

The final report treats the Kalman filter as a separate, small preprocessing block. The intended implementation is a simple 1-D Kalman filter for the soil-moisture signal. Its job is to reduce noise and spikes and provide a steadier moisture input. It is a preprocessing aid, not a new agricultural model and not a safety authority.

3.4 Layer 4 - ESP32 connectivity

The ESP32 is the communication bridge. It receives the processed telemetry from the STM32 over UART, connects to Wi-Fi, sends data to the Node.js/Express backend, receives configuration/advisory information, retries after network interruptions, and can buffer telemetry when offline. It is not the local pump safety authority.

4. Final Hardware Design and Responsibilities

4.1 Controllers

Component	Responsibility	Why it stays
STM32F103C8T6	Sensor acquisition, local safety, state machine, pump control, feedback	Real-time embedded authority and offline safety
ESP32	UART bridge, Wi-Fi, telemetry, configuration, reconnect	Keeps connectivity separate from physical control

4.2 Final core sensors

Sensor	Interface / signal	What TerraFlow uses it for
Capacitive moisture	Analog ADC	Primary irrigation condition; calibrated moisture
DS18B20	1-Wire	Soil temperature and context
TDS sensor	Analog ADC	Water-quality / dissolved-solids context
BME280	I2C	Temperature, RH, pressure; VPD/WSI/DRI context
BH1750	I2C	Illuminance / crop and environmental context

4.3 Final added measurement and safety hardware

- NEO-6M GPS: zone location and weather API lookup.
- Flow sensor: mandatory in the final implemented system for measured water delivery.
- Float switch: mandatory tank-low protection.
- AUTO / OFF / MANUAL mode selection: clear human control and safe demonstration.
- STM32 watchdog: firmware recovery to a safe state.
- MOSFET driver with gate resistor, pull-down and flyback protection: safe low-voltage pump switching.

4.4 Power and actuator path

Final physical actuation path

```
Control: STM32 GPIO / PWM -> gate resistor -> IRLZ44N MOSFET
Pull-down: gate -> 10k -> GND
Pump path: supply -> DC pump -> MOSFET -> GND
Protection: flyback diode across pump
Measurement: pump outlet -> flow sensor -> irrigation line -> Z01
```

5. Final Software Architecture and Data Flow

5.1 Retained software stack

```
STM32 -> UART -> ESP32 -> Wi-Fi -> Node.js / Express -> Supabase -> React
                  |
                  +-> Weather API context
                  +-> Agronomic knowledge stored in Supabase
                  +-> Decision engines + ML support
```

The project intentionally retains Node.js/Express, Supabase and React rather than rewriting the working infrastructure merely to use a different protocol. The ESP32 can communicate through HTTP/HTTPS; MQTT is not a required migration target for the final project.

5.2 Operational data vs reference knowledge

Supabase contains two logical categories of information. Operational data are the changing observations and events: sensor time series, irrigation events, zone/device state, soil-test entries and advisories. The Agronomic Knowledge Base is reference/configuration data: crop profiles, soil profiles, crop stages, seasons, irrigation rules, fertilizer rules and their sources. Both live in Supabase but have different purposes.

5.3 Weather context

BME280 and the Weather API are complementary. BME280 reports the current local state near TF01. The Weather API provides external forecast context such as rainfall probability/forecast, temperature, humidity, wind and ET0 when available. The two are not blindly averaged into a synthetic sensor reading. The decision engine sees current local state and forecast context as separate evidence.

5.4 Derived agro-environmental context

- VPD: derived from local temperature and relative humidity; used as atmospheric-demand context.
- WSI and DRI: retained as supporting indices for interpretation and dashboard display.
- ET0: reference evapotranspiration context from valid weather data when available.
- ETc = ET0 x Kc: crop water-demand context using the crop/stage Kc stored in the knowledge base.

> **Important boundary**
>
> ET0/Kc are used as contextual agricultural variables. TerraFlow does not claim that its five continuous sensors alone produce a rigorously validated FAO-56 ET0 or a full crop-growth simulation.

6. Agronomic Knowledge Base and Dataset Design

The knowledge base is intentionally small. The goal is not to encode every agricultural fact; it is to store enough structured information for consistent, explainable decisions in Z01.

6.1 Soil profiles

Profile	Primary role
Black	Water-behaviour context, crop suitability, fertilizer context
Red	Water-behaviour context, crop suitability, fertilizer

Profile	Primary role
	context
Alluvial / Muddy	Water-behaviour context, crop suitability, fertilizer context
Sandy / White	Water-behaviour context, crop suitability, fertilizer context

These are software/reference profiles. They do not mean the live sensors directly measure every soil property in the profile.

6.2 Crop library

Detailed stage-wise logic	Basic suitability coverage
Rice	Cotton
Wheat	Soybean
Maize	Groundnut
Tomato	Chickpea
	Onion
	Potato

6.3 Growth stages and seasons

Growth stage	Meaning in TerraFlow
Initial	Early establishment and starting water-demand context
Development	Demand generally increases as crop develops
Mid-season	Often a key high-demand / sensitivity period
Late-season	Demand changes as crop approaches maturity

Seasons stored in the final knowledge base: Kharif, Rabi and Zaid.

6.4 Source and provenance rule

Every agronomic record should carry a source or methodology identifier. This matters because an examiner may ask where a Kc value, crop suitability range or fertilizer dose came from. The answer should point back to the stored source rather than a hardcoded number with no traceability.

7. Irrigation Decision and Water Delivery Logic

This is the most important technical section

7.1 Baseline vs final context-aware controller

Level	Logic	Purpose
Baseline	Calibrated moisture below configured threshold -> irrigation candidate	Proven simple reference controller and fallback
TerraFlow	Soil + crop + stage + calibrated moisture + weather/rain + ET0/Kc + zone area + safety	Context-aware decision support and target-volume calculation

The baseline is not deleted. It is retained so the final evaluation can compare a simple reference against the richer TerraFlow logic.

7.2 Final irrigation sequence

1. Get the latest valid zone state: calibrated moisture, crop, stage, soil profile, weather context and relevant pump history.
2. Estimate crop water demand context using ETc = ET0 x Kc where valid, and assess the current moisture deficit.
3. Determine an irrigation depth / target water amount for the zone using the configured agronomic logic and the known zone area.
4. Convert target irrigation depth into target water volume in litres.
5. Read measured flow rate from the flow sensor and estimate the required runtime.
6. Before starting, STM32 checks tank level, operating mode, sensor validity, runtime limits and other local safety conditions.
7. Run the pump.
8. Accumulate delivered volume from measured flow rather than using a fixed-duration assumption.
9. Stop when the target volume is reached, or earlier if a fault / timeout / safety condition occurs.
10. Log the irrigation event and perform post-irrigation moisture feedback verification.

7.3 Water quantity equations

Core equations - keep them simple and explainable

```
ETc = ET0 x Kc
Target volume (L) = Target irrigation depth (mm) x Zone area (m^2)
Estimated runtime (min) = Target volume (L) / Measured flow rate (L/min)
Delivered volume = time integration of measured flow (or calibrated pulse-to-volume conversion)
```

The target irrigation depth is not a universal constant. It is derived from the selected crop/stage/soil context, current moisture state, weather/rain context and the configured control method.

7.4 Why there is no fixed 5-minute irrigation cycle

The system can still monitor sensors periodically, but irrigation itself is event/need-based. Once an event starts, runtime is determined by the target water volume and measured flow. This avoids the earlier contradiction of saying both "fixed 5-minute irrigation" and "volume-based delivery".

Final control statement

Sensors may be monitored periodically. The pump is not forced to run on a fixed 5-minute cycle. The actual irrigation event ends when target volume is reached or a safety/timeout condition stops it.

8. Crop Suitability Advisory

The crop module is a pre-planting or seasonal advisory. It evaluates the ten configured crop profiles against the current zone conditions and returns the top five recommendations with suitability scores and reasons.

8.1 Inputs

- Selected soil profile.
- Season: Kharif, Rabi or Zaid.
- Current environmental context: temperature and light, with water availability/moisture context.
- Weather context when relevant.
- Crop profile requirements stored in the knowledge base.

8.2 Hybrid method

The transparent core remains rule/range based: each candidate crop is compared against soil, season, temperature, water and other available requirements. The separate ML support layer can provide an additional score or prediction where the training data justify it. The architecture therefore remains hybrid rather than forcing every decision to be made by a black-box model.

8.3 Final output

```
Top 5 crop recommendations
Suitability score / category
Short justification: matched conditions and limiting conditions
Current zone, season and environmental context
```

> **Important distinction**
>
> A higher suitability score does not automatically mean higher profit. Profitability requires economic inputs such as expected yield, production cost and market price, which are outside the locked core project.

9. Fertilizer Advisory

Fertilizer is a separate advisory path. The system accepts soil-test/reference nutrient values and combines them with crop, growth stage, season and soil context. The five live sensors do not directly measure N, P and K.

9.1 Final fertilizer input path

```
Soil test / reference case: N, P, K, pH, EC (and OC where available)
                +
Crop + growth stage + season + soil profile
                |
          Recommendation method
                |
           Fertilizer advisory
                |
      N / P2O5 / K2O + reason + source
```

9.2 Agronomic method

The project uses documented soil-test-based methods and reference rules. Depending on the selected reference method, the recommendation may be stage-wise and may be expressed per hectare or for the defined zone area. The exact coefficients and doses must be populated from the chosen reference sources during dataset construction.

9.3 ML role - deliberately left for analysis

The final architecture keeps a hybrid pathway, but the exact fertilizer ML model is not assumed to be solved before validation. A valid ML model requires appropriate labelled data, meaningful features and an evaluation procedure. Therefore the rule-based advisory is mandatory, while ML support is an evidence-driven work package.

> **Do not say this in viva**
>
> "Our TDS sensor tells us the NPK level." That would be incorrect. The correct statement is: "TDS is used as irrigation-water dissolved-solids context; fertilizer uses soil-test/reference N/P/K inputs."

10. Machine Learning Decision Support

ML is a separate support layer, not the owner of the pump. This distinction is central to the final architecture.

10.1 Final ML position

Module	Primary logic	ML role
Irrigation	Rule-based context assessment	Random Forest support / prediction; evaluated against baseline
Crop suitability	Transparent rule-based scoring	Optional / supporting hybrid score where validated
Fertilizer	Soil-test rule/recommendation method	Model to be analyzed and validated before being treated as evidence

10.2 Feature discipline

Potential ML features include calibrated moisture, soil temperature, BME280 environment, light, TDS context, weather variables, crop, growth stage and other zone metadata. The model should only use features that are actually available, correctly calibrated or sourced, and represented consistently in the training data.

10.3 What happens when ML disagrees with rules?

The final project treats ML as support. The decision engine should preserve explicit agronomic rules, reason codes and local safety conditions. An ML score can strengthen or weaken confidence in a recommendation, but it should not bypass STM32 safety checks or create a hidden cloud-only control path.

10.4 No LSTM/GRU requirement

LSTM/GRU is not a mandatory final deliverable. The supplied roadmap explicitly keeps deep temporal models outside the core scope so that effort remains focused on calibration, controller evaluation and explainable agricultural logic.

> **ML success criterion**
>
> A smaller model with clean features, honest validation and clear limitations is more valuable for this project than a complex model with weak or fabricated evidence.

11. Data Quality, Reason Codes and Explainability

The data-quality layer is intentionally lightweight. Its job is to detect stale, missing, invalid or suspicious inputs before they enter the decision logic.

11.1 Checks

- Freshness: is the reading recent enough for the decision?
- Range: is the value physically / sensor-wise plausible?
- Missingness: is a required value absent?
- Stuck sensor: has the value remained unchanged suspiciously long?
- Weather availability: is the forecast/context present and fresh?
- Configuration completeness: are crop, stage, soil and zone settings present?

11.2 Reason codes

Code	Meaning
RUN: MOISTURE_LOW	Calibrated moisture indicates irrigation need
RUN: WATER_DEFICIT	Context supports irrigation after demand/moisture assessment
SKIP: MOISTURE_ADEQUATE	No immediate irrigation need
SKIP: RAIN_CONTEXT	Forecast rain context reduces immediate need
FAULT: SENSOR_INVALID	Unsafe or invalid sensor input
FAULT: TANK_LOW	Float switch blocks irrigation
FAULT: NO_IRRIGATION_RESPONSE	Expected post-irrigation moisture response not observed

11.3 Confidence language

Use HIGH, MEDIUM or LOW evidence status instead of presenting a made-up calibrated probability. HIGH can mean complete, fresh inputs with known crop/stage/soil/weather. MEDIUM or LOW can indicate missing or stale evidence. A model probability should only be shown as a probability if the project later demonstrates that it is calibrated.

12. Database Design

Supabase is the persistent data layer. The Agronomic Knowledge Base is inside Supabase, represented by multiple structured tables rather than one giant table.

Table	Purpose	Typical fields
zones	Defines the current zone	zone_id, farm_id, device_id, area, crop, soil_type, season, stage
sensor_data	Stores time-series observations	timestamp, zone_id, moisture, soil_temp, tds, air_temp, humidity, pressure, lux, latitude, longitude, flow
crop_profiles	Crop suitability and requirement data	crop, season, soil_preferences, pH context, temperature range, water requirement, sensitivity, light context
soil_profiles	Software/reference soil context	soil_id, soil_name, water_behavior, pH range, EC/salinity context
crop_stages	Stage-wise water context	crop, stage, duration, water_demand, stress_sensitivity, Kc
irrigation_rules	Context rules	soil, crop, stage, moisture condition, weather, rain condition, action, reason, source
fertilizer_rules	N/P/K advisory logic	crop, stage, season, N status, P status, K status, recommendation, method, source
soil_tests	Entered/reference soil-test cases	zone_id, date, N, P, K, pH, EC, OC where available
irrigation_events	Pump event history	zone_id, start, stop, runtime, target volume, delivered volume, reason, feedback
advisories	Human-visible recommendations	zone_id, type, recommendation, reason, confidence, timestamp, model version

12.1 Why configuration is stored in the database

The decision engine should not contain hundreds of hardcoded crop and soil constants. When the values are stored as configuration/reference data, the same logic can be reused for another crop or zone by changing the data rather than rewriting firmware or backend code.

13. Firmware State Machine and Safety

Recommended STM32 state machine



```
        STOP        CONTINUE
          |
    RE-SENSE / FEEDBACK
```

13.1 Safety gates

Check	Action if failed
Mode = OFF	Pump OFF
Tank level low	Pump OFF + fault/notification
Sensor invalid	Pump OFF / safe fallback
Maximum runtime exceeded	Pump OFF + fault
Flow abnormal or absent	Stop or raise fault based on configured rule
Network unavailable	Core local safety/irrigation remains independent

13.2 Watchdog

The STM32 independent watchdog should return the controller to a safe state if firmware execution becomes stuck. The pump hardware should default to OFF during reset, supported by the MOSFET gate pull-down arrangement.

13.3 Manual modes

- AUTO: normal TerraFlow decision and safety path.
- OFF: pump always disabled.
- MANUAL: controlled operator test mode, still bounded by hardware safety where appropriate.

14. Phase 1 - Completed Embedded Irrigation Foundation

Phase 1 established the physical control loop before the cloud and ML layers were added. The five core sensors were acquired by STM32, the pump was switched through a MOSFET stage, and post-irrigation moisture was re-read to verify the response.

14.1 Completed items

Component	Phase 1 result
STM32F103C8T6	Sensor acquisition, filtering/scaling, local evaluation, GPIO control
Five sensors	Moisture, DS18B20, TDS, BME280, BH1750 acquired
Pump driver	GPIO -> MOSFET -> low-voltage pump
Local irrigation	Preliminary threshold logic with local control
Feedback	Moisture re-read after irrigation

14.2 What Phase 1 proved

- The controller can acquire the sensor streams.

- Local irrigation can run without cloud dependency.
- The pump can be controlled from STM32 hardware.
- The loop is closed because the system measures the response after actuation.

Phase 1 message for viva

First we made the physical system work. We did not start with AI; we first established sensing, local control and feedback.

15. Phase 2 - Completed Connected Prototype

Phase 2 extended the embedded foundation into an end-to-end software system. The existing telemetry, backend, database, dashboard, WSI/DRI, Random Forest support and basic crop scoring are treated as the completed connected prototype.

Area	Completed capability
Telemetry	UART sensor payloads and Wi-Fi forwarding
Backend	Ingestion, latest/history, prediction/pump paths
Supabase	Sensor/state/ML/crop persistence
WSI / DRI / VPD	Derived agro-environmental context
Random Forest	IRRIGATE / NO IRRIGATE decision support
Crop scorer	Basic range-based suitability scoring
React	Live values, trends, indices, pump state and crop result
Closed-loop UI	Pump action and feedback visible end-to-end

ESP32 consistency rule

Older project material may mention ESP-01S. The final locked connectivity device is ESP32. Slides, diagrams and the final report should use ESP32 consistently.

15.1 What Phase 2 proved

- The embedded node and digital stack can work together end-to-end.
- Sensor data can be stored, analyzed and visualized.
- ML can assist irrigation decisions without owning the pump.
- A crop recommendation pipeline already exists and can be expanded into the final knowledge base.

16. Phase 3 - Agronomic Intelligence and Validation

DATASET -> CALIBRATION -> DECISION LOGIC -> INTEGRATION -> VALIDATION

Phase 3 is the main current-semester software and agronomy phase. Its purpose is to convert a connected sensor prototype into a crop-, stage-, soil- and weather-aware system without turning the project into a large simulation platform.

16.1 Work package sequence

11. Freeze the 4 soil profiles and 10 crop profiles.
12. Build 16 detailed crop-stage records for Rice, Wheat, Maize and Tomato across four growth stages.
13. Build sourced fertilizer reference rules for the four detailed crops.
14. Calibrate the capacitive moisture sensor for the physical demonstration soil.
15. Integrate the 1-D Kalman filter as a simple soil-moisture preprocessing block.
16. Integrate weather forecast and ET0/Kc context.
17. Upgrade irrigation logic to context-aware target-volume assessment.
18. Implement the top-5 crop suitability advisory.
19. Implement the soil-test fertilizer advisory and analyze ML support.
20. Add reason codes and qualitative evidence status.
21. Evaluate Random Forest as decision support against the simpler baseline.
22. Validate the complete advisory + local actuation loop.

16.2 Phase 3 completion criteria

- Knowledge base populated and stored in Supabase.
- Moisture calibration complete for the selected demonstration setup.
- Rice, Wheat, Maize and Tomato have detailed stage-wise water context.
- All ten crops have basic suitability profiles.
- Fertilizer advisory works for the four detailed crops.
- Weather and ET0/Kc context is integrated where valid.
- Random Forest is evaluated as decision support.
- Reason codes and data-quality states are visible.
- The STM32 automatic pump loop remains safe and functional.

17. Phase 4 - Hardware Hardening, Measurement and Deployment

Make TF01 stable, measurable and defensible

Phase 4 is not a second AI project. It is the reliability and measurement phase that turns the working prototype into a more stable final demonstration and gives the project stronger experimental evidence.

17.1 Final Phase 4 hardware scope

Task	Purpose
Enclosure / cable organization	Protect the node and make the field setup repeatable
Power / electrical protection	Stable low-voltage operation and motor-noise control
Float switch	Block dry-run when tank is low
AUTO / OFF / MANUAL	Clear operation and safe demonstration
Watchdog	Recover from firmware lock-up
Flow sensor	Measure actual water delivery and close the volume loop
Fault detection	Detect no-flow, overruntime, tank-low and sensor problems
Offline / reconnect	Do not lose local safety when Wi-Fi fails
Long-duration test	Show repeatability and stability

Task	Purpose
Multi-zone architecture	Keep software scalable without building multiple permanent nodes

17.2 Final hardware list - nothing else

The locked core implementation contains the capacitive moisture sensor, DS18B20, TDS, BME280, BH1750, NEO-6M GPS, STM32F103C8T6, ESP32, DC pump + MOSFET driver, float switch and flow sensor. Optional future additions are not part of the final core build.

17.3 Phase 4 exit criteria

- TF01 is physically organized and protected.
- Pump and power path are verified.
- Float, AUTO/OFF/MANUAL and watchdog functions work.
- Flow is measured and irrigation events record delivered volume.
- Fault conditions are detected and logged.
- Network loss does not remove local safety.
- Long-duration behavior is tested.
- The software remains structured for future multi-zone replication.

18. Testing and Experimental Validation Plan

The project must distinguish real measured experiments from software validation cases and reference datasets. Only measured results should be presented as measured results.

18.1 Experiment 1 - Moisture sensor calibration

Step	Action	Evidence
1	Collect sensor readings under several repeatable moisture conditions	Raw sensor table
2	Record a simple reference moisture measurement	Reference values
3	Fit a practical curve or lookup relationship	Calibration equation / table
4	Check repeatability and operating range	Error / repeatability summary
5	Store calibrated thresholds as configuration	Controller settings

18.2 Experiment 2 - Controller comparison

Compare three approaches: fixed schedule, simple calibrated moisture threshold, and TerraFlow context-aware controller. The main metrics are pump runtime, number of irrigation events, water delivered, moisture trajectory and recovery after irrigation.

18.3 Experiment 3 - Zone representativeness

Use the permanent TF01 sensor point plus a small number of temporary reference points inside Z01. Calculate mean, spread, coefficient of variation and node-vs-reference difference. The purpose is to test the reasonableness of the chosen point, not to claim whole-farm coverage.

18.4 Experiment 4 - Crop suitability

Create compatible and incompatible scenarios and verify that the rule-based score moves in the expected direction. This is a software validation case, not a yield experiment.

18.5 Experiment 5 - Fertilizer advisory

Use representative or available soil-test cases and compare the TerraFlow recommendation with the chosen documented method. Label it as a reference/software case study unless an actual field experiment is performed.

18.6 Irrigation control metrics

Metric	Meaning
Target water volume (L)	Planned amount for the event
Delivered water volume (L)	Measured amount from flow sensor
Volume error (%)	Difference between target and measured delivery
Pump runtime	Time needed to deliver the volume
Moisture change	Pre- vs post-irrigation response
Event count	How often irrigation is initiated
No-response faults	Events where expected moisture response is not observed

19. Expected Outputs and Final Demonstration

19.1 Physical demonstration

The most important live demo

```
Dry / low moisture condition
-> irrigation assessment required
-> STM32 safety checks
-> pump ON
-> measured water flow
-> target volume reached
-> pump OFF
-> moisture rises
-> feedback verified
```

19.2 Dashboard views

- Zone Z01 live monitoring: moisture, soil temperature, TDS, BME280 values, light, GPS/location and pump state.
- Irrigation page: target volume, delivered volume, runtime, reason code, safety status and feedback.
- Crop page: top five recommendations with suitability score and justification.

- Fertilizer page: soil-test values, crop/stage, recommendation, method and source.
- History page: irrigation events, trends and previous advisories.
- Data-quality status: fresh/valid context and any current fault or evidence limitation.

19.3 Example explanation flow

What the examiner should see

```
Zone: Z01
Crop: Tomato
Stage: Mid-season
Moisture: LOW (calibrated)
Weather: rainfall not sufficient for current decision window
Water status: DEFICIT
Irrigation: REQUIRED
Target volume: calculated for zone area
Flow: measured in real time
STM32: safety checks passed -> PUMP ON
Stop condition: target volume reached
Feedback: moisture response verified
```

> **The "wow" point**
>
> The project is not just a sensor dashboard. It closes the loop from physical sensing to a reasoned decision to physical water delivery and then back to measured verification.

20. Limitations, Non-Claims and Future Scope

20.1 Current limitations

- One physical node and one physical zone do not establish whole-farm spatial coverage.
- The capacitive moisture reading needs practical calibration for the demonstration soil.
- TDS provides water-quality context and is not NPK.
- Crop suitability and fertilizer quality depend on the quality and provenance of the configured knowledge base.
- The Random Forest model depends on its training data, labels and features.
- The ten-crop library is mostly a software knowledge base; not all ten crops are physically validated.
- Weather API availability and freshness affect contextual recommendations.
- A multi-season, multi-location agricultural trial is outside the current major-project scope.

20.2 Features deliberately excluded from the core

- LSTM / GRU as a mandatory model
- AquaCrop or full crop-growth simulation
- Satellite / NDVI integration
- Multiple permanent nodes
- Complex GIS / spatial modeling
- Direct NPK sensor architecture
- Unnecessary MQTT / WebSocket migration
- Large-scale commercial farm management claims

20.3 Future scope

- Replicate the TF01 design into multiple management zones.

- Expand the crop and soil knowledge base after validating the core cases.
- Use larger real datasets to improve ML support.
- Add advanced crop-water models or satellite context if the project later becomes a research program.
- Perform multi-season and multi-location validation.
- Add a separate economic layer if profitability estimation is ever required.

21. Research Paper Positioning

The project can support one strong paper before attempting a second. The most natural first paper is the embedded closed-loop irrigation study because it can be backed by real measurements from the physical system.

21.1 Paper 1 - recommended focus

Low-Cost Embedded Closed-Loop Irrigation with Sensor Calibration and Controller Evaluation

- Capacitive moisture calibration.
- STM32 local control.
- Target-volume irrigation with flow measurement.
- Fixed schedule vs threshold vs TerraFlow context-aware comparison.
- Moisture response and water-use metrics.
- Reason-coded safety and feedback.

21.2 Paper 2 - conditional

Zone-Aware Agro-Environmental Decision Support for Crop Suitability and Soil-Test-Based Fertilizer Advisory. This second paper becomes stronger only if the crop and fertilizer modules have enough data and validation evidence.

> **Research rule**
>
> Build the core for real. Use clearly labelled reference datasets and software validation cases where necessary. Never present unperformed experiments or invented numerical results as measured evidence.

22. Viva-Ready Explanations

Q: Why STM32 if there is a backend?

A: STM32 is the local safety and actuation authority. The backend can recommend and provide context, but pump control must remain safe and locally bounded.

Q: Why is the Kalman filter separate?

A: It is a small preprocessing stage for the soil-moisture signal. It reduces noise and makes the input more stable for decision logic.

Q: Why is the flow sensor necessary?

A: It converts irrigation from estimated runtime into measured water delivery. The controller can target a water volume and stop when the measured volume is reached.

Q: Why not calculate runtime only from pipe radius?

A: Pipe radius gives cross-sectional area, not actual flow. Actual delivery also depends on flow velocity and pump behavior, so measured flow is more direct for the final control loop.

Q: Why no fixed 5-minute cycle?

A: Monitoring may be periodic, but the irrigation event is need-based. Runtime depends on the target volume and measured flow.

Q: Is TDS an NPK sensor?

A: No. TDS indicates dissolved-solids concentration in the irrigation water and is used only as water-quality context.

Q: Where is the Agronomic Knowledge Base?

A: Inside Supabase, as structured tables such as crop_profiles, soil_profiles, crop_stages, irrigation_rules and fertilizer_rules.

Q: What is the role of ML?

A: Decision support. ML helps the decision engines, but it does not directly control the pump.

Q: Does crop suitability predict profit?

A: No. It ranks crop suitability from the configured agronomic and environmental evidence. Profit estimation would require economic data.

Q: How does fertilizer work without NPK sensing?

A: The module accepts soil-test/reference N/P/K values, combines them with crop, stage, season and soil context, and applies a documented recommendation method.

Q: Does one node cover the whole farm?

A: No. It represents the defined management zone Z01. Whole-farm coverage would require multiple zones and spatial validation.

Q: Why keep the simple threshold controller?

A: It is a baseline and fallback. Keeping it lets us compare simple local control with the enhanced TerraFlow context-aware controller.

Q: What happens when Wi-Fi fails?

A: The embedded safety and basic local irrigation path remains independent of the cloud. Connectivity can reconnect and buffered telemetry can be forwarded later.

Q: Why not use LSTM?

A: The current dataset and project scope do not justify it as a mandatory model. Random Forest and structured features are enough for the intended decision-support study.

Appendix A. Final Locked Lists

A.1 Hardware

Category	Locked items
Core sensors	Capacitive moisture; DS18B20; TDS; BME280; BH1750
Location	NEO-6M GPS
Water / safety	Flow sensor; float switch
Controllers	STM32F103C8T6; ESP32
Actuation	Low-voltage DC pump; MOSFET driver; flyback protection; gate resistor; gate pull-down
Human interface	AUTO / OFF / MANUAL; LEDs; buzzer
Reliability	STM32 watchdog; fault handling

A.2 Software

Layer	Locked software
Embedded	STM32 firmware; local safety; pump state machine; feedback
Preprocessing	1-D soil-moisture Kalman filter
Connectivity	ESP32 UART + Wi-Fi
Backend	Node.js / Express
Database	Supabase
Application	React dashboard
Agricultural logic	Irrigation; crop suitability; fertilizer advisory
Context	Weather API; ET0/Kc; VPD; WSI; DRI
ML	Random Forest decision support; hybrid support framework

A.3 Knowledge base

- 4 soil profiles.
- 10 crop profiles.
- 16 detailed crop-stage records for 4 core crops x 4 stages.
- 3 seasons.
- Irrigation context rules.
- Fertilizer rules with source/method traceability.

Appendix B. Pseudocode

B.1 Final irrigation control pseudocode

```
LOOP:
  read sensors
  validate freshness and ranges
  filter soil moisture
  read local weather + forecast context
  load zone / soil / crop / stage configuration
  assess irrigation need
  if decision != REQUIRED:
     keep pump OFF
     log reason
     continue to next monitoring cycle
  calculate target irrigation depth
  target_volume_L = target_depth_mm * zone_area_m2
  measure flow rate
  estimate runtime
  STM32 safety checks
  if any safety check fails:
     pump OFF; log fault; notify
  else:
     pump ON
     while delivered_volume < target_volume and no fault:
```

```
        integrate flow pulses / measured flow
        enforce maximum runtime
    pump OFF
    log irrigation event
    re-read moisture after irrigation
    evaluate response
```

B.2 Final crop suitability pseudocode

```
for crop in crop_profiles:
    compare season
    compare soil context
    compare temperature / environment
    compare moisture / water availability
    compare light where relevant
    compute transparent suitability score
    store reasons
rank by suitability score
return top 5 + reasons
```

B.3 Final fertilizer advisory pseudocode

```
read soil_test N, P, K, pH, EC, OC if available
read crop + stage + season + soil profile
load fertilizer method / rule
calculate or retrieve N / P2O5 / K2O recommendation
attach reason + source
optionally add validated ML support output
return advisory
```

Appendix C. Glossary and Quick Revision

Term	Simple meaning
ADC	Converts an analog sensor voltage into a number the MCU can read
UART	Serial communication link between STM32 and ESP32
I2C	Shared two-wire bus used by BME280 and BH1750
1-Wire	Digital communication method used by DS18B20
Kalman filter	A simple estimator used here to smooth the noisy moisture signal
Calibration	Converts the raw sensor response into a usable project-specific moisture value
VPD	A measure of the atmosphere's drying demand based on temperature and RH
ET0	Reference evapotranspiration
ETc	Crop evapotranspiration / crop water-demand context
Kc	Crop coefficient used with ET0 to estimate ETc

Term	Simple meaning
TDS	Total dissolved solids; water-quality context, not NPK
WSI	Water stress index used as supporting agro-environmental context
DRI	Disease-risk environmental proxy used as supporting context
RF	Random Forest machine-learning model
Zone	A physically defined area managed together; here Z01
Node	One TerraFlow electronic unit; here TF01
Reason code	Short label describing why a decision was made
Target volume	The amount of water the irrigation event is intended to deliver
Flow rate	Water delivery rate measured by the flow sensor
Feedback	Checking the physical response after irrigation

Final 10-line revision

1) STM32 owns safety and actuation. 2) ESP32 owns connectivity. 3) Kalman filters soil moisture. 4) Supabase stores operational data and the agronomic knowledge base. 5) Weather API adds forecast context. 6) ML supports decisions. 7) Irrigation uses target water volume, not a fixed 5-minute run. 8) Flow sensor measures delivered volume. 9) Crop returns top 5 suitability recommendations. 10) Fertilizer uses soil-test/reference N/P/K and is not derived from TDS.

Source Basis. Documents used to construct this final consistency report

Primary source documents supplied in the conversation were used as the baseline: the Final Phase 1-4 Project Roadmap & Execution Plan, the Detailed Major Project Report, and the Complete Project Guide. The latest architecture image supplied by the project team was used as the visual reference. Where older source documents conflicted with the later locked decisions, the later locked decisions were used for this final consistency version, especially the mandatory flow sensor, distinct Kalman filter block, separate ML support layer, Supabase Agronomic Knowledge Base, event/need-based target-volume irrigation, top-five crop recommendations, and hybrid fertilizer architecture.

Important document discipline

- Do not insert measured numbers into this report unless the team has actually collected them.
- Replace any remaining placeholder agronomic values with cited values from the selected agricultural references.
- Label reference data, software validation cases and demonstration scenarios honestly.
- Keep the architecture, firmware, dashboard, database and viva wording synchronized with the locked scope.

Final submission checklist

Check	Final requirement
Architecture	One node TF01 / one zone Z01; Kalman filter is a distinct block; ML is a separate support layer
Hardware	Five core sensors + GPS + flow sensor + float switch + STM32 + ESP32 + protected pump path
Irrigation	Need-based target-volume control; no fixed 5-minute irrigation cycle
Water delivery	Zone area used for target volume; flow sensor used for measured delivery and runtime
Database	Operational data and Agronomic Knowledge Base stored in Supabase
Crop advisory	Top 5 suitability recommendations with score and justification
Fertilizer	Soil-test/reference N/P/K; rule-based core; ML support only after evidence
Safety	STM32 local checks, float protection, watchdog, runtime limit, safe pump OFF state
Evidence	Measured experiments separated from reference data and software validation cases
Claims	No whole-farm coverage claim, no NPK-sensor claim, no automatic profit claim

Final sign-off principle

From this point onward, slides, code, database, dashboard, report and viva answers should use the same terminology and the same control boundary. Any new feature must be explicitly marked as future scope before it enters the architecture.

Prepared as the project team's final technical understanding/reporting baseline. It is designed to be readable for learning and viva preparation as well as for formal project documentation.